

Foundations of Programming

Assignment No. : 4

1. **PROBLEM STATEMENT:** Write a C Program to calculate the salary of an employee given his basic pay (taken as input from the user). Calculate gross salary of employee. Let HRA be 10 % of basic pay and TA be 5% of basic pay. Let employees pay professional tax as 2% of total salary. Calculate net salary payable after deductions.

OBJECTIVE:

1. To understand the importance of flowchart for any programming model.
2. To learn simple flowchart symbols and arrows to define relationships.
3. To understand and develop visual representations of flow of data.

THEORY:

1. Definition of Algorithm

An **algorithm** is a finite, step-by-step procedure used to solve a specific problem or perform a computation. It provides a clear sequence of instructions that start from given input, process the data logically, and produce the desired output within a limited number of steps.

2. Characteristics of an Algorithm

1. **Input** – An algorithm accepts zero or more inputs.
 2. **Output** – It produces at least one result or output.
 3. **Definiteness** – Each step must be clear and unambiguous.
 4. **Finiteness** – The algorithm must terminate after a finite number of steps.
 5. **Effectiveness** – Every step should be basic and executable in practice.
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3. Theory: Employee Salary Calculation

In payroll systems, the salary of an employee is calculated using different components such as **basic pay, allowances, and deductions**.

- **HRA (House Rent Allowance)** is provided as a percentage of basic pay to support housing expenses.
- **TA (Travel Allowance)** is given to cover transportation costs.
- The **gross salary** is the sum of basic pay and all allowances.
- Certain deductions like **professional tax** are applied on the total salary.
- After subtracting deductions from gross salary, the **net salary payable** is obtained.

This type of calculation demonstrates the use of arithmetic operations and user input handling in C programming.

ALGORITHM:

Step 1: Start

Step 2: Declare variables for basic pay, HRA, TA, gross salary, tax, and net salary

Step 3: Read basic pay from the user

Step 4: Calculate

HRA = 10% of basic pay

TA = 5% of basic pay

Step 5: Compute gross salary

Gross Salary = Basic Pay + HRA + TA

Step 6: Calculate professional tax

Tax = 2% of gross salary

Step 7: Compute net salary

Net Salary = Gross Salary – Tax

Step 8: Display gross salary and net salary

Step 9: Stop

CODE:

```
#include <stdio.h>
```

```
int main() {
```

```
    float basic, hra, ta, gross, tax, net;
```

```
    printf("Enter basic pay: ");
```

```
scanf("%f", &basic);

hra = 0.10 * basic;

ta = 0.05 * basic;

gross = basic + hra + ta;

tax = 0.02 * gross;

net = gross - tax;

printf("\nGross Salary = %.2f", gross);

printf("\nNet Salary = %.2f", net);

return 0;

}
```

```

1  #include <stdio.h>
2
3  int main() {
4      float basic, hra, ta, gross, tax, net;
5
6      printf("Enter basic pay: ");
7      scanf("%f", &basic);
8
9      hra = 0.10 * basic;
10     ta  = 0.05 * basic;
11
12     gross = basic + hra + ta;
13     tax   = 0.02 * gross;
14
15     net = gross - tax;
16
17     printf("\nGross Salary = %.2f", gross);
18     printf("\nNet Salary   = %.2f", net);
19
20     return 0;
21 }
22 |

```

PLATFORM: Linux

INPUT: 20,000

OUTPUT:

```

Enter basic pay: 20000

Gross Salary = 23000.00
Net Salary   = 22540.00

```

CONCLUSION: Thus, we have learned to write algorithms and to draw its representation in terms of flowchart.

FAQs:

1. Describe various symbols and their use in flowchart.
2. Difference between Algorithm and Flowchart.